

Subject Description Form

Subject Code	EE2901S
Subject Title	Basic Electricity and Electronics
Credit Value	3
Level	2
Pre-requisite/ Co-requisite/ Exclusion	Nil
Objectives	1. To introduce students to the fundamental principles and analysis techniques for studying and operating basic electrical and electronic devices and circuits. 2. To introduce students to the appropriate skills and tools for experimenting with basic electrical and electronic devices and circuits.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Describe the fundamental principles and analysis techniques for studying and operating basic electrical and electronic devices and circuits. b. Use the appropriate skills and tools for experimenting with basic electrical and electronic devices and circuits.
Subject Synopsis/ Indicative Syllabus	DC Electrical Circuits Electromechanical analogies. Common Prefixes. How electricity works. Atoms and atomic structure. Basic electrical quantities: charge, potential, current, voltage, power, and energy. Resistance, Ohm's law, and resistors. Resistors in series and in parallel. Sign convention. Practical, ideal, independent, and dependent voltage and current sources. Voltage and current dividers. Use of basic test meters: voltmeters, ammeters, ohmmeters, and multimeters. Lumped circuit elements. Network description: branch, node, loop, and mesh. Kirchhoff's voltage and current laws. Tellegen's theorem. Mesh-current and node-voltage methods. Thévenin's and Norton's theorems. Loading effect and maximum power transfer. AC Electrical Circuits Tesla vs. Edison: the war of the currents. Differences between AC and DC. AC signal basics: time-dependent, periodic, and sinusoidal signals and sources. Worldwide mains electricity. Peak, average, and root-mean-square values. Inductors and capacitors. Sinusoidal steady-state analysis by time-domain method. Phasor and impedance basics: complex numbers, Euler's identity, phasor representation, impedance and admittance, phasor diagrams. Sinusoidal steady-state analysis by phasor-impedance method. Power, energy, and electricity bill. Instantaneous and average powers. Power in resistive, inductive, capacitive, and complex loads. Complex power and power factor. Power generation, transmission, and distribution. Three-phase power basics. Diodes and Diode Circuits Semiconductor basics: basic electrical and electronic materials, energy band diagrams, intrinsic and extrinsic semiconductors, electrons and holes, doping, donors and acceptors, n-type and p-type semiconductors. P-N junction diodes: basic structure, symbol, depletion region, barrier potential, forward bias, reverse bias, ideal current-voltage characteristics, ON and OFF states, ideal diode equation, breakdown characteristics, Zener diodes. Diode circuits: ideal and practical diode assumptions, basic and specific diode circuits.

	Bipolar Junction Transistors (BJTs) and BJT Amplifiers BJTs: basic structures, symbols, BJT semiconductor basics, BJTs as electric switches and amplifiers. DC biasing and circuits: modes of operation, DC biasing, input and output characteristics, DC equivalent circuits, DC analysis, load line, Q-point, DC biasing schemes. AC small signal and amplifiers: linear analog amplifiers, notation, basic characteristics of BJT amplifiers, DC and AC analyses with equivalent circuits, small-signal hybrid-π equivalent circuits with and without early effect, small-signal amplifier parameters, common-emitter amplifiers, DC and AC load lines. Laboratory Experiments: - E01: DC Circuits. - E02: Diode Circuits.																				
Teaching/Learning Methodology	<table><tr><th rowspan="2">Teaching/Learning Methodology</th><th colspan="2">Outcome</th></tr><tr><th>a</th><th>b</th></tr><tr><td>Lecture</td><td>✓</td><td></td></tr><tr><td>In-class Practice (Tutorial)</td><td>✓</td><td></td></tr><tr><td>Assignment</td><td>✓</td><td></td></tr><tr><td>Test</td><td>✓</td><td></td></tr><tr><td>Laboratory</td><td></td><td>✓</td></tr></table> Lecture: Students are introduced to the knowledge of the subject, and their comprehension is strengthened with interactive Q&A (outcome a). In-class Practice (Tutorial): Students apply what they have learned to solve the problems in the class (outcome a). Assignment: Students further test and develop their understanding and comprehension of the knowledge through after-class exercises (outcome a). Test: Students develop their actual understanding and comprehension of the knowledge (outcome a). Laboratory: Students acquire hands-on/simulated experience using the appropriate electrical and electronic devices, test equipment, and design software in circuits, apply what they have learned in the class to experimentally validate the theoretical investigations, and develop the experimental log and report writing skills (outcome b).	Teaching/Learning Methodology	Outcome		a	b	Lecture	✓		In-class Practice (Tutorial)	✓		Assignment	✓		Test	✓		Laboratory		✓
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Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="2">Intended learning outcomes to be assessed</th></tr><tr><th>a</th><th>b</th></tr><tr><td>Continuous Assessment</td><td>50%</td><td>✓</td><td>✓</td></tr><tr><td>Examination</td><td>50%</td><td>✓</td><td></td></tr><tr><td>Total</td><td>100%</td><td colspan="2"></td></tr></table> Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: Overall Assessment: $0.5 \times \text{Continuous Assessment} + 0.5 \times \text{End of Subject Examination}$	Specific assessment methods/tasks	% weighting	Intended learning outcomes to be assessed		a	b	Continuous Assessment	50%	✓	✓	Examination	50%	✓		Total	100%				
Specific assessment methods/tasks	% weighting			Intended learning outcomes to be assessed																	
		a	b																		
Continuous Assessment	50%	✓	✓																		
Examination	50%	✓																			
Total	100%																				

	Continuous Assessment covers all intended learning outcomes a and b, while Examination involves intended learning outcome a. Continuous Assessment (50%) contains Assignment (18%), Test (18%), and Laboratory (14%), aiming to provide timely feedback to students on various topics of the syllabus, including assignment works, laboratory skills, use of the appropriate electrical and electronic devices, test equipment, and design software in circuits, analysis and presentation of the experimental results, etc. Examination (50%) is a three-hour, closed-book, end-of-subject written examination, aiming to assess students' overall understanding and ability to apply the fundamental principles and analysis techniques.	
Student Study Effort Expected	Class contact:	
	▪ Lecture	28 Hrs.
	▪ In-class Practice (Tutorial)	3 Hrs.
	▪ Test	2 Hrs.
	▪ Laboratory	6 Hrs.
	Other student study efforts:	
	▪ Self-study	44 Hrs.
	▪ Assignment	12 Hrs.
	▪ Laboratory Log and Report	10 Hrs.
	Total student study effort	105 Hrs.
Reading List and References	Textbooks: 1. Giorgio Rizzoni and James Kearns, <i>Principles and Applications of Electrical Engineering</i>, 7th Edition, New York: McGraw Hill (2022). 2. Donald A. Neamen, <i>Microelectronics: Circuit Analysis and Design</i>, 6th Edition, New York: McGraw Hill (2016). Reference books: 1. William H. Hayt, Jack E. Kemmerly, Jamie D. Phillips, and Steven M. Durbin, <i>Engineering Circuit Analysis</i>, 10th Edition, New York: McGraw Hill (2024). 2. Alan H. Robbins and Wilhelm C. Miller, <i>Circuit Analysis: Theory and Practice</i>, 5th Edition, New York: Cengage Learning (2013). 3. Raymond A. DeCarlo and Pen-Min Lin, <i>Linear Circuit Analysis: Time Domain, Phasor, and Laplace Transform Approaches</i>, 2nd Edition, New York: Oxford University Press (2001).	